

Name:

Form:

MYP 1 Year 7 Chemical Reactions

Statement of
Inquiry:

Observing chemical reactions gives **evidence** to help understand **patterns** and **relationships** in the world around us.

Key Concept:

Relationships

Related
Concepts:

Patterns and Evidence



7C MYP Command Terms

Term	Definition
Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.
Apply	Use knowledge and understanding in response to a given situation or real circumstances
Describe	Give a detailed account or picture of a situation, event, pattern or process
Design	Produce a plan, simulation or model
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence
Document	Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography
Evaluate	Make an appraisal by weighing up the strengths and limitations
Explain	Give a detailed account
Formulate	Express precisely and systematically the relevant concept(s) or argument(s)
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
Outline	Give a brief account
Present	Offer for display, observation, examination or consideration
Recall	Remember or recognize from prior learning experiences
Select	Choose from a list or group
Solve	Obtain the answer(s) using appropriate methods
State	Give a specific name, value or other brief answer without explanation or calculation
Suggest	Propose a solution, hypothesis or other possible answer
Summarize	Abstract a general theme or major point(s)

Glossary: Chemical Reactions

Word	Meaning
Chemical reaction	A change where new chemicals are formed
Irreversible change	A change in which what you have made cannot be turned straight back into what you started with
Physical change	A change where no new chemical is formed
Reversible change	A change where what you have made is easily turned back into what you started with
Reactant	Chemicals that join together to make a new chemical
Product	A new chemical formed in a chemical reaction
Observation	Something you see or smell when doing a reaction
Hydrogen	A flammable gas that can be made by chemical reactions
Corrosive	Chemicals that attack other chemicals and skin causing damage
Carbon Dioxide	A gas that puts out fires and can be made by chemical reactions
Limewater	A chemical that is used to test whether a gas is carbon dioxide
Fuel	A chemical that can release energy when it is reacting
Hydrocarbon	A chemical that only contains hydrogen and carbon
Flammable	Can catch fire and burn easily
Combustion	The chemical reaction when we burn a fuel

Acid	A chemical that turns litmus red. It has a pH of less than 7
Alkali	A chemical that turns litmus blue. It has a pH of more than 7
Hazardous	A chemical that can cause damage
Precaution	Something we do to protect ourselves from hazards in an experiment (like wearing goggles)
Corrosive	Chemicals that attack metals, rocks and skin very quickly
Irritant	Chemicals that make the skin and eyes itchy

Neutral	Chemicals that are not acids or alkalis. They have a pH of 7
Litmus	A simple indicator. It goes red in acids, blue in alkalis and purple in neutrals

Indicator	A chemical that changes colour when it is in an acid and alkali
Extract	To remove a chemical from a plant so it can be used

pH Scale	A measurement from 1 - 14 that shows how strong acids and alkalis are.
Universal Indicator	A mixture of indicators that gives a range of colours showing how strong or weak acids and alkalis are

Neutralisation	A reaction adding acid to an alkali to form a neutral chemical
Solution	A mixture of water and another chemical that is dissolved into it

7C 1a Is there a physical change or chemical change?

Apparatus

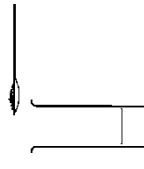
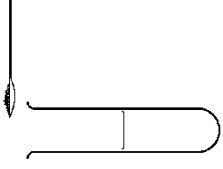
- Bunsen burner, Goggles, Heatproof mat, Spatula, Test tubes, Test tube holder, Tripod, Test tube rack, Chemicals

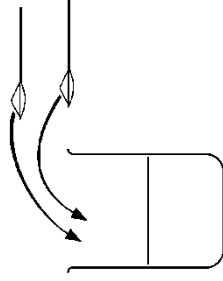
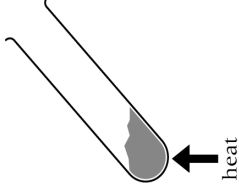
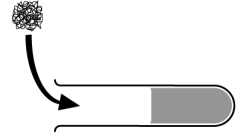
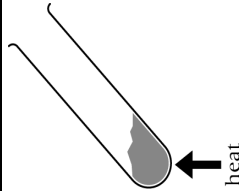
Safety

Wear goggles. Treat all chemicals as irritants. Be careful when heating chemicals. Your teacher will show you how to safely heat experiment 4 and 6.

Method

- Carry out the experiments described in the table, and look carefully at what happens.
- Write your observations in the table. Use the 'What happened?' column.
- In the right column of the table say if you think a physical or chemical change has taken place.
 Chemical change - a new substance is formed and the change is not easily reversed (physical changes are easily reversed).

Experiment	What to do	Diagram	What happened?	Is it a physical or chemical change?
1	- Half fill a test tube with lemon juice. - Add a spatula of bicarbonate of soda. - Watch what happens. - Feel the tube.		Fizzed and bubbled (a gas was given off); the tube felt cold.	Chemical change
Chemical change - the starting material changed into a new substance (gas given off) and the change cannot easily be reversed.				
2	- Half fill a small test tube with water. - Add a spatula of baking powder. - Watch what happens. - Feel the tube.		Fizzed and gave off bubbles; the tube felt cold.	Chemical change

3	- Put 20cm³ of water into a small beaker with water. - Add two spatulas of salt. - Stir the mixture thoroughly. - Heat the beaker over a Bunsen until the water has all boiled away.		The water boiled away and white salt crystals were left behind.	Physical change
4	- Put two spatulas of zinc oxide in a test tube. - Heat the tube. Watch what happens. - Leave the hot tube in a rack to cool down.		The white powder turned yellow when it was heated.	Physical change
5	- Half fill a test tube with copper sulphate solution. - Put a little iron wool into the copper sulphate solution. - Watch what happens.		The blue solution went pale green and brown copper coated the iron wool.	Chemical change
6	- Put a few pieces of wax onto a test tube - Heat the test tube with a Bunsen flame. - Allow to cool back down and write down what you see.		The wax melted to a clear liquid, then set solid again as it cooled.	Physical change

Conclusion

How can you tell from these observations if a chemical or physical change is taking place?

A chemical change makes new chemicals and cannot easily be reversed (gas, colour change, temperature change). A physical change makes no new chemical and is easily reversed.

Can you give some examples of things to look for to tell if a chemical change is taking place?

Look for bubbles or fizzing (gas given off), a colour change, a new solid, a change in temperature or a new smell.

7C 1b Reactant or Product

Look at the following examples, so of which you completed in the practical (sheet 1a) try to put these chemicals into the correct group - were they a reactant or product?

E.g. Iron wool + copper sulfate $\longrightarrow$ iron sulfate + copper

Citric acid + Bicarbonate of soda $\longrightarrow$ sodium carbonate + carbon dioxide + water

Magnesium + Oxygen $\longrightarrow$ Magnesium oxide

Reactants are the starting chemicals; products are the new substances formed. Sort each chemical into the correct column.

Reactant	Product
Reactants: iron wool, copper sulfate, citric acid, bicarbonate of soda, magnesium, oxygen	Products: iron sulfate, copper, sodium carbonate, carbon dioxide, water, magnesium oxide

Conclusion

What do you think we mean by a 'reactant' or 'product' in a chemical reaction?

Reactants are the chemicals that join together at the start of a reaction; products are the new chemicals made in the reaction.

Do both chemical changes and physical changes have reactant and products?

No. Only chemical changes have reactants and products, because new chemicals are made. Physical changes make no new chemical.

What do you think the arrow means in a chemical reaction? Reactant $\longrightarrow$ Product

The arrow means 'changes into' (reacts to make) - it points from the reactants to the products.

Extension

Research and try to find the chemical formulas for some of your reactants and products

E.g. magnesium Mg + oxygen O₂ · magnesium oxide MgO; iron Fe + copper sulfate CuSO₄ · iron sulfate FeSO₄ + copper Cu.

7C 1c Chemical or Physical Change

- 1** Define a physical change – (use the keyword: reversible)

A physical change is a change where no new chemical is formed and it is reversible (it can easily be changed back).

- 2** Define a chemical change – (use the keywords: permanent irreversible)

A chemical change is a change where new chemicals are formed. It is permanent and irreversible.

- 3** Give three examples of how you know a chemical change has taken place – use the

- 2** Identify whether the example is a chemical or physical change.

Bubbles of gas are given off (fizzing), the colour changes, and the temperature changes.

- a** If you put some water into a freezer, it turns to ice. Physical change
- b** Bananas turn black if you leave them in a warm cupboard for a week. Chemical change
- c** Some white powder from your kitchen fizzes when you put it into vinegar. Chemical change
- d** When you boil water in a kettle, some of it turns to steam. Physical change
- e** When you fry an egg it turns white and goes solid. Chemical change

Extension: For one chemical change and one physical change above (Q2), can you explain why you made your decision.

Physical: ice melting is reversible and no new chemical is made. Chemical: the egg white changes colour and goes solid and cannot be changed back, so a new chemical has formed.

7C 2a Types of Chemical Reactions – Making Hydrogen

How can you collect the gas given off in an experiment?

Apparatus

- Goggles
- Test tubes x 3 and stoppers
- Small Conical flask (use instead of a boiling tube shown in the diagram)
- Washing up bowl
- Bung and delivery tube
- 3 pieces of magnesium (1cm long each)
- Measuring cylinder
- Dilute hydrochloric acid
- Bunsen burner
- Splints

Safety:

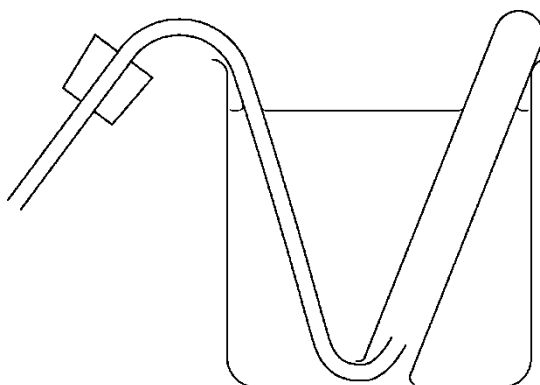
Hydrogen is an explosive gas.
Only produce small amounts to be tested in a test tube.
Pack away the equipment **before** testing the H_2 .
Hydrochloric acid is an irritant.



Wear eye protection.

Method

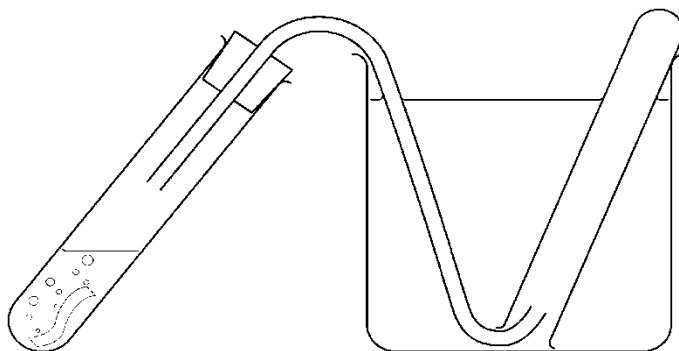
- 1 Fill the 3 test tubes and the washing up bowl with water. Put your thumb over the end of the test tube, and put it into the washing up bowl upside down.
- 2 Put the end of the delivery tube into the upside down test tube, without letting any of the water out.



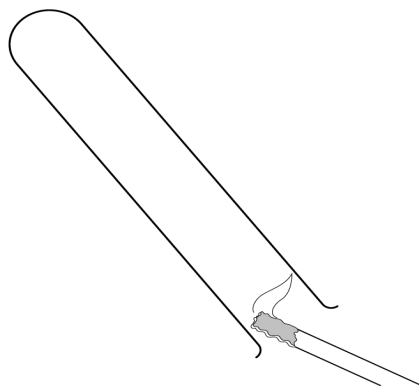
- 3 Measure 25 cm³ of hydrochloric acid (using a measuring cylinder) and put it into a small conical flask.

Bubble the gas into a test tube full of water held upside down in a bowl of water - the gas pushes the water out.

- 4 Drop 3 pieces of magnesium into the acid, and put the bung into the top of the flask. It is better to use a small conical flask rather than the boiling tube shown in the following diagram.



- 5 Watch what happens to the magnesium and the acid.
- 6 When the first test tube is filled, shake any water out and put a stopper in, try and collect 3 test tubes full of gas.
- 7 When the reaction has finished, pack away your equipment **before** you test your gas.
- 8 Carefully test the gas in the upside down tube using a lighted splint. Your teacher will show you how to do this safely. Point the test tube away from yourself and others.



Considering your results/conclusions

- 1 What are the names of the two reactants in this experiment? Magnesium and hydrochloric acid
- 2 What did you see happening when you put the magnesium into the acid?
The magnesium fizzed and bubbles of gas were given off; the metal got smaller.
- 3 How could you tell that one of the products of this reaction is a gas?
Bubbles of gas were given off and the gas collected in the upside-down test tube, pushing the water out.

4 What happened when you tested the gas in the test tube? The lighted splint went out with a squeaky pop.

5 What does this tell you about the gas?
It tells us that the gas is hydrogen, because hydrogen burns with a squeaky pop with a lighted splint.

Extension:

Find out the chemical formula of hydrogen gas. What is the product when it reacts with oxygen during this test? Hydrogen gas is H_2 . It reacts with oxygen to make water: $2H_2 + O_2 \rightarrow 2H_2O$.

7C 2b Reactions of Metals with Acids

- 1 How can you tell from the diagram that a reaction is happening in the test tube?

Bubbles of gas are being given off from the metal.



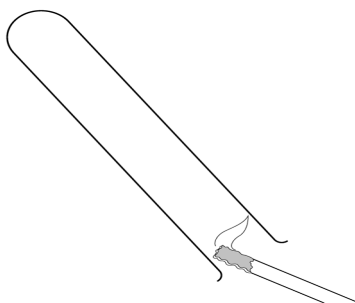
If you looked at the thermometer while the reaction was happening, what would you expect to see?

The temperature would rise (the mixture would get warmer).

What safety precautions should you take if you did this experiment?

Wear goggles and be careful with the acid; use a small amount of acid and a test tube rack.

- 2 Describe how you would test a gas to see if it is hydrogen.



Collect the gas in an upside-down test tube (over water).

Put a lighted splint into the mouth of the test tube.

If the gas is hydrogen, what would you see? A squeaky pop

Extension:

1. Rainwater is a weak acid.

a) What can rainwater do to metals like iron? The weak acid reacts with the iron and corrodes (rusts) it.

b) What types of things are made out of metals like iron?

Bridges, cars, ships, gates, railings, tools and cutlery.

c) Why do you think that it is a disadvantage to make things out of iron?

Iron rusts (corrodes) in rainwater, so things made from it become weak and damaged and must be replaced.

d) What other materials can be used instead?

Plastics, aluminium, stainless steel, glass, wood or carbon fibre.

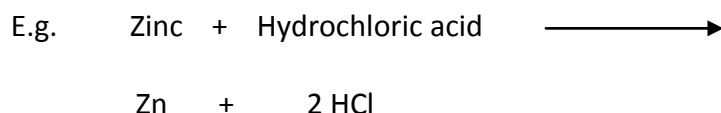
2. a) Why do some metals, such as gold not react with acids? Can you think of any other metals that may not react? You may need to research this.

Gold is very unreactive, so it does not react with acids. Other unreactive metals are platinum and silver.

b) How does the reactivity of the metal affect what we use it for? E.g. Why do we use gold for jewelry?

Unreactive metals stay shiny and do not corrode, so we use gold for jewellery; reactive metals like iron are used where strength matters.

c) Can you find the chemical equation that describes when a metal such as zinc reacts with hydrochloric acid. Zinc + hydrochloric acid → zinc chloride + hydrogen; $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$.



7C 3a Making Carbon Dioxide

How can we find out which materials contain carbonates?

Carbon dioxide is formed when an acid is added to a carbonate. Carbon dioxide is a gas. You can see that a gas is being made because bubbles are produced and the gas produced puts out a lit splint

Apparatus

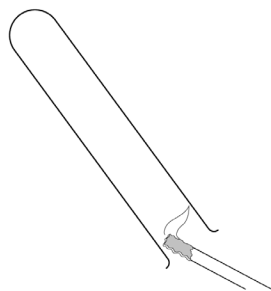
- Goggles
- Boiling tubes and a delivery tube
- Test tube rack
- Hydrochloric acid
- Materials to test



Wear eye protection.

Method

- 1 Collect the materials you are going to test and write their names in the table.
- 2 Add about 1 cm depth of acid to the boiling tube.
- 3 Put half a spatula of a material into the tube.
- 4 Watch carefully and write your observations in the table.
- 5 If a gas is given off, collect some of the gas by connecting a delivery tube and a test tube. Your teacher will show you how to do this.
- 6 Put a **lit splint** into the test tube, write down what you observe.



- 6 Repeat steps 2 to 5 for your other materials.

Recording your results

Material	Observation	What happened to the lit splint?	Is the material a carbonate?
A squeaky 'pop' is heard - this shows hydrogen gas was produced.			
Marble Chips	There was a steady stream of small bubbles released.	The lit splint went out.	Yes
Chalk	Steady bubbles of gas; the chalk slowly dissolved.	The lit splint went out.	Yes
Bicarbonate of soda	Fizzed quickly with lots of bubbles.	The lit splint went out.	Yes
Iron (nail)	A few bubbles formed slowly and the metal dissolved.	There was a squeaky pop.	No
Salt	No bubbles - the salt just dissolved.	Nothing happened - no gas was collected.	No
Limestone	Bubbles of gas were given off.	The lit splint went out.	Yes

Your teacher will demonstrate how to test the gas formed to prove it is carbon dioxide using limewater.

Draw a labelled diagram of the equipment your teacher used to test the gas.

· Labelled diagram: boiling tube holding the material and acid, delivery tube leading into a test tube of limewater; labels: boiling tube, acid, material, delivery tube, limewater, gas bubbles.

You can now test one of your materials in the same way using limewater.

If the gas made is carbon dioxide what did you observe?

The limewater turned milky (cloudy white).

Considering your results/conclusion

How did you know which materials contained carbonates?

The materials that fizzed and gave off gas when acid was added, and whose gas turned limewater milky, contained carbonates.

How could you prove that the gas produced was carbon dioxide rather than another gas?

Bubble the gas through limewater - if it turns milky/cloudy white the gas is carbon dioxide.

Is carbon dioxide heavier or lighter than air? How can you tell?

Heavier than air - it collects at the bottom of the tube and can be poured onto a flame to put it out.

What is the chemical formula of carbon dioxide? What does this formula tell you about the elements that it contains?

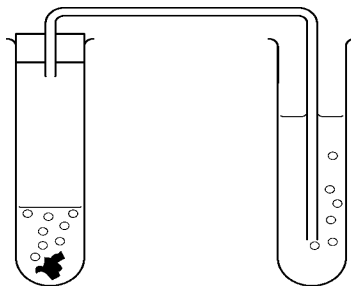
CO₂ - the formula tells you it is made of carbon and oxygen (1 carbon atom and 2 oxygen atoms).

Extension:

Research some more things about carbon dioxide gas, for example, how is it related to combustion or respiration? Why is it used to make carbonated (fizzy) drinks or in fire extinguishers?

CO₂ is made by burning fuels and by respiration and is used by plants. It is used in fizzy drinks and in fire extinguishers because it does not burn and smothers flames.

7C 3b Properties of Carbon Dioxide and Hydrogen Gas



- 1 Describe how you would test a gas to see if it is carbon dioxide.

Bubble the gas through limewater; if the limewater turns milky the gas is carbon dioxide.

If the gas *is* carbon dioxide, what would you see?

The limewater turns milky (cloudy white).

- 2 Which gas would you get if you carried out these reactions?

hydrochloric acid + magnesium produces ~~hydrogen~~..... gas

nitric acid + calcium carbonate produces ~~carbon dioxide~~.. gas

hydrochloric acid + magnesium carbonate produces ~~carbon dioxide~~.. gas

sulphuric acid + magnesium produces ~~hydrogen~~..... gas

- 3 Think about the tests you have completed for hydrogen and carbon dioxide gas. List the similarities in tests and difference in the tests.

Similarities between tests	Differences between tests
Both gases are collected in a test tube and both are tested by putting a lighted splint into the tube.	Hydrogen burns with a squeaky pop; carbon dioxide puts the splint out. Carbon dioxide is tested with limewater, hydrogen with a lighted splint.

Extension:

Hydrogen can be used as fuel. Explain the advantages and disadvantages of this.
Research how carbon dioxide fire extinguishers work.

7C 3c Gases in Air – Extension Activity

1. The names of some gases found in the air are listed below. Write the names of the gas in the left-hand column of the table next to the correct set of properties. You may need to look up the gases in a science text book or on the internet.

argon carbon dioxide helium methane nitrogen
nitrogen dioxide oxygen

Gas	Properties
carbon dioxide	This gas is in the air we breathe out. It puts flames out. It turns limewater milky.
oxygen	This is the gas we need to stay alive. It is also needed for burning. A burning splint will burn brightly in this gas.
nitrogen dioxide	This is a brown gas with a strong smell. It is poisonous. It can cause acid rain.
nitrogen	This gas makes up most of the air on Earth. It does not react very much.
methane	This is the gas in the gas taps. It burns with a blue flame if there is plenty of air.
argon	This gas does not react at all. It is used in light bulbs so that the metal in the light bulb can get very hot without burning.
helium	This gas is very light, so we use it in balloons. It does not react at all, and will put out a flame. Divers use this gas.

- 2 The next table shows how much of each gas there is in the air.

Gas	Amount in the air
nitrogen	78%
oxygen	21%
argon	0.9%
other gases (including helium, methane and carbon dioxide)	0.1%

Draw a pie chart to show the composition of the air. You can draw it by hand using a compass and protractor, or you can use a computer program to plot the pie chart for you. Colour in the sections on your pie chart and provide a key.

· Pie chart with key: nitrogen 78%, oxygen 21%, argon 0.9%, other gases 0.1%, each section coloured and labelled.

7C 4a Types of Chemical Reaction - Burning Metals

Copper is a metal which reacts with the oxygen in the air when it is heated strongly.

Prediction

I think that if I weigh some copper powder and then heat it strongly in the air, the mass of the metal will go up (stay the same/go up/go down).

I think this because The copper reacts with the oxygen in the air when it is heated.

The oxygen atoms join onto the copper atoms to make copper oxide.

The extra oxygen adds mass, so the copper oxide weighs more than

the copper did at the start - nothing has been lost.

Apparatus

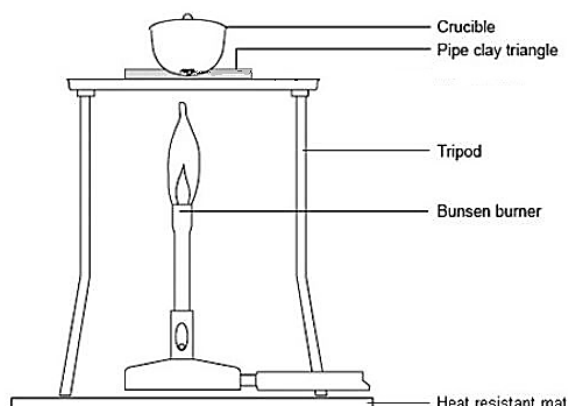
- Bunsen burner and tripod.
- Heat-proof mat
- Crucible and pipe triangle.
- Copper powder
- Goggles and tongs.



Wear eye protection. Be careful with hot apparatus

Method

- 1 Zero the balance then add a spatula of copper powder to a crucible and weigh them accurately. Record the mass in the table.
- 2 Look carefully at the surface of the copper and write down what it looks like in the table.
- 3 Put your crucible on a tripod using a pipe triangle like the diagram.



- 4 Heat the crucible strongly in the flame of a Bunsen burner for about 3 minutes. Write down your observations in the table. Is this a chemical or a physical change?
- 5 After 3 minutes, turn off the Bunsen and let the crucible **cool down for at least 5 minutes**. Write down what the reacted copper powder looks like after heating.
- 6 When it is cool, use the tongs to carefully lift the crucible onto the heat proof mat. Your teacher will show you how to do this. Take the mat and the crucible over to the balance and record the mass of the crucible. Write this in your table.

Chemical change - new substances were formed and the change is not easily reversed.

Recording your results

Complete the table to show your results.

	Mass shown on balance (g)	Observation
Copper powder and crucible before heating	Shiny orange-brown copper powder.	20.50
Copper powder and crucible during heating	The copper glows and turns black as it reacts.	Not measured - the crucible was being heated.
Reacted Copper powder and crucible after heating	The powder is black (copper oxide).	20.72

Calculation: How can you work out if the copper has gained or lost mass during heating?

Write down your working and your answer in the space below.

Take away the mass before heating from the mass after heating (mass after - mass before).

$20.72\text{ g} - 20.50\text{ g} = +0.22\text{ g}$, so the copper has gained 0.22 g of mass.

Considering your results/conclusion

From my results I can see that after I had heated the copper metal the mass had

gone up (gone up/stayed the same/gone down).

I think this happened because: (Hint: What does the copper react with when it burns?)

The copper reacted with oxygen from the air to make copper oxide.

The oxygen atoms joined onto the copper, adding extra mass.

Nothing escaped from the crucible, so the mass went up.

Complete the word equation: Copper + Oxygen $\longrightarrow$ Copper oxide

Extension $\rightarrow$

Complete the symbol equation: $2\text{Cu} + \text{O}_2 \longrightarrow 2\text{CuO}$

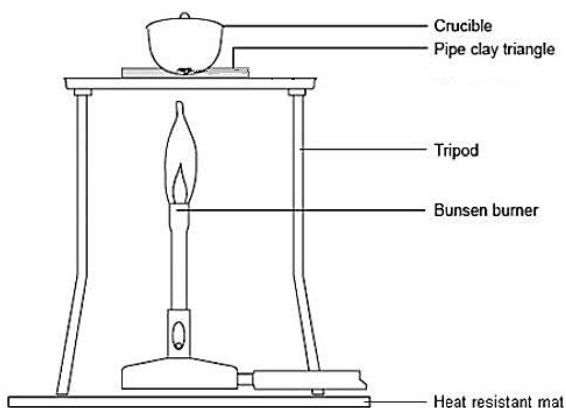
If 2.5g of copper is heated with 0.5g of oxygen, how much copper oxide would you expect to get? 3.0 g, because the total mass stays the same in a reaction: $2.5\text{ g} + 0.5\text{ g} = 3.0\text{ g}$.

What do you think conservation of mass means?

The total mass of the reactants equals the total mass of the products - mass is not created or destroyed in a reaction.

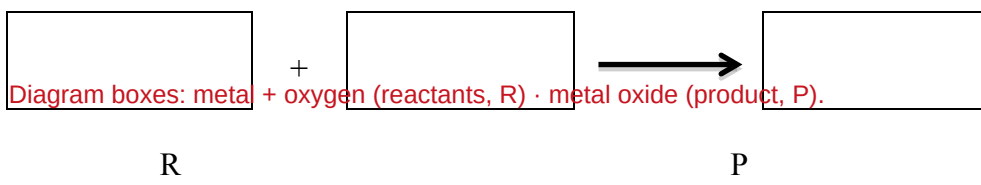
7C 4b Word Equations – Burning Metals

1 When metal burns, a chemical reaction happens.



Fill in the boxes and the gaps in the sentences, using words from the list.

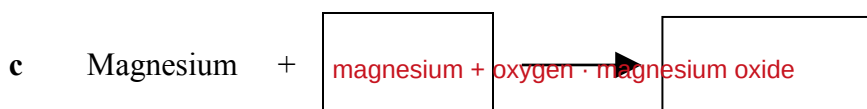
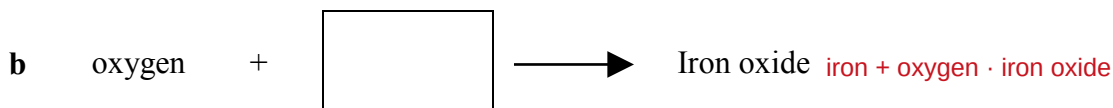
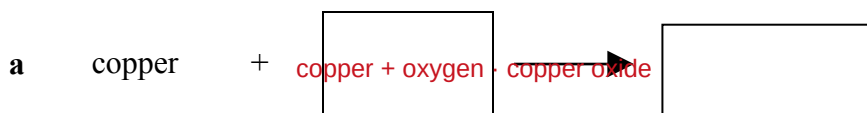
Metal, products, combustion, air, reactants, fuel, irreversible, oxygen, metal oxide



This type of reaction is called a combustion reaction. It is irreversible.

The metal is the fuel and oxygen comes from the air.

2 Now complete these word equations for the combustion of different materials.



Extension: When a metal reacts to form a metal oxide, the mass increases. Why?

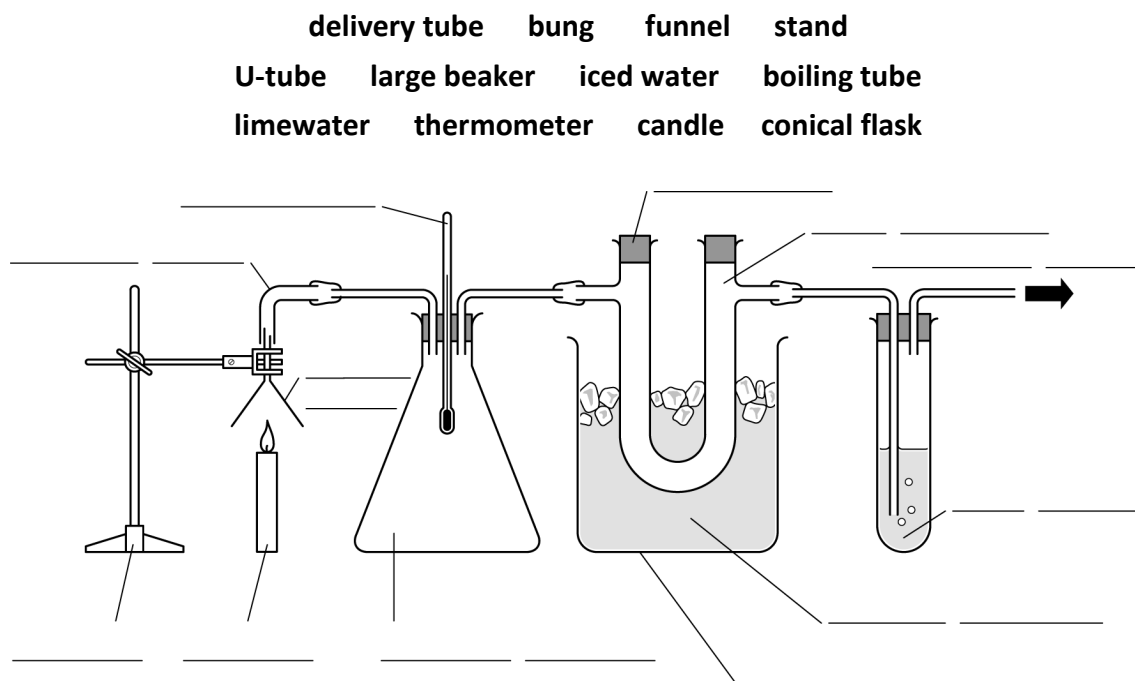
Because the metal atoms join with oxygen atoms from the air, so the oxygen adds extra mass to the metal oxide.

7C 5a Types of Chemical Reaction - Combustion of Fuels

What happens when fuels burn?

Your teacher will demonstrate the combustion of a fuel. Label the diagram using words from the list below.

· Labels: candle; large beaker; funnel; delivery tube; U-tube in iced water; boiling tube with limewater; thermometer; stand; bung.



Recording your results

Now write the results from the demonstration in the table below:

Observation	Before experiment	After experiment
Appearance of fuel	A white solid candle (wax)	The candle is shorter - some wax has been used up
Appearance of limewater	Colourless (clear) liquid	Milky (cloudy white)
Appearance of U-tube	Empty and dry	Colourless liquid (water) collected in it
Temperature shown on thermometer (°C)	About 20 °C (room temperature)	Higher than before the experiment

How do you know that a chemical change has taken place?

New substances were made - the limewater turned milky (carbon dioxide) and water collected in the U-tube; the change is irreversible.

Write the word equation for the reaction: fuel (candle wax) + oxygen → carbon dioxide + water

7C 5b Types of Chemical Reaction - Combustion of Fuels - 2

Prediction

I think that if I weigh a candle and then burn it in the air, the mass of the candle will

go down (stay the same/go up/go down).

I think this because The wax reacts with oxygen from the air when it burns (combustion)

Carbon dioxide gas and water vapour are made and they escape into

the air, so they are not weighed. This means the candle has

less mass at the end than at the start.

Apparatus

- Lighter
- Heat-proof mat
- Candle
- Goggles and tongs.



Wear eye protection. Be careful with hot apparatus

Method

- 1 Zero the balance then weigh your candle accurately (to all three decimal places). Record the mass in the table.
- 2 Look carefully at the candle and write down what it looks like in the table.
- 3 Put your candle on a heatproof mat and ensure it will not fall over.
- 4 Light the candle and leave it burning for about 3 minutes. Write down your observations in the table. Is this a chemical or a physical change?
- 5 After 3 minutes, blow the candle out carefully. **Wait for the wax on the top to cool down and become solid again.** Write down what the candle looks like after burning.
- 6 When it is cool, take the mat and the candle over to the balance and record the mass of the candle. Write this in your table.

Chemical change - a new product was formed (e.g. gas given off) and the change is not easily reversed.

Recording your results

Complete the table to show your results.

	Mass shown on balance (g)	Observation
Candle before combustion	A white solid candle with a wick.	45.620
Candle during combustion	The wax melted and burned with a yellow flame; wax ran down the candle.	Not measured - the candle was burning.
Candle after combustion	The candle is shorter and the melted wax at the top has set solid again.	44.980

Calculation:

How can you work out if the candle has gained or lost mass during heating?

Take away the mass after burning from the mass before burning (mass before - mass after).

Write down your working and your answer in the space below.

45.620 g - 44.980 g = 0.640 g, so the candle has lost 0.640 g of mass.

Considering your results/conclusion

From my results I can see that after I had burnt the candle the mass had gone down

I think this happened because: (Hint: What does the candle produce when it burns?)

The wax reacts with oxygen from the air when the candle burns.

Carbon dioxide and water vapour are made and escape into the air, so

they are not weighed - the candle has lost mass.

Complete the word equation:

Candle wax + oxygen $\longrightarrow$ carbon dioxide + water + energy

Extension:

Find out what 'phlogiston' is and how this influenced our understanding of the conservation of mass and burning.

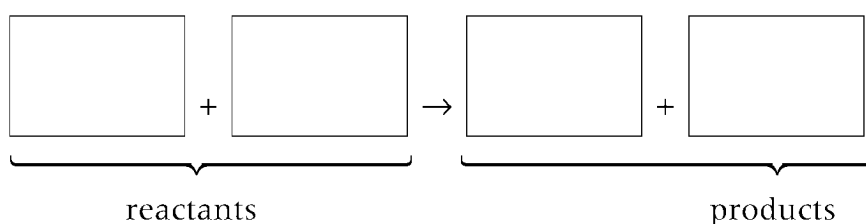
Phlogiston was an old idea that a burning substance releases a fire-like material. It was wrong - burning metals actually gain mass as they join with oxygen.

7C 5c Chemistry of a Candle Burning

1. When a candle burns, a chemical reaction happens.



Fill in the boxes using words from the list.



candle wax carbon dioxide oxygen water

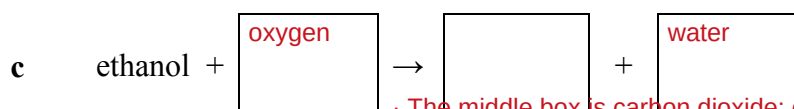
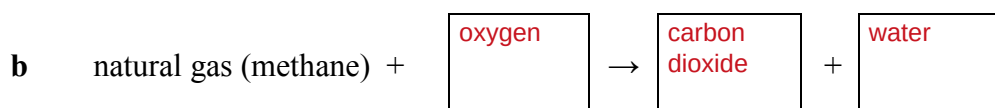
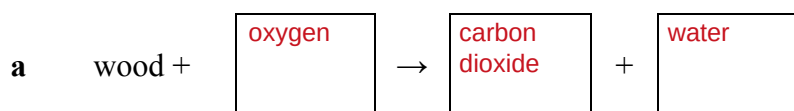
· Diagram boxes: candle wax + oxygen (reactants) · carbon dioxide + water (products).

2. Write a description of what is happening as the candle is burning. Try and use as many of the words below as possible

Reactants combustion fuel irreversible air oxygen products water
carbon dioxide energy

The candle wax is the fuel (reactant) that reacts with oxygen in the air. This combustion is irreversible and makes the products carbon dioxide and water, releasing energy.

3. Now complete these word equations for the combustion of different materials.



· The middle box is carbon dioxide: ethanol + oxygen · carbon dioxide + water.

7C 6a What Hazard?

Match the warning symbol to the hazard, and find the correct example.

Colour in or number them or draw lines to match them up.

Give the hazard for each substance, dilute sulphuric acid, toxic, flammable, liquid, harmful if swallowed.



Hazard:
flammable

Example:
concentrated
sulphuric acid



Example:
dilute sulphuric acid

Hazard:
irritant

Hazard:
toxic



Example:
petrol

Example:
mercury

Hazard:
corrosive



Extension:

Find out what other hazard symbols there are that are not included here.

Explosive, oxidising, harmful/health hazard, serious health hazard, gas under pressure, dangerous for the environment.

Find out some examples of chemicals that would need to be labeled with these symbols.

Explosive: fireworks; oxidising: hydrogen peroxide; environmental hazard: bleach; gas under pressure: butane lighter fuel.

Why is it important to have hazard symbols in Chemistry?

So people can see how dangerous a chemical is before using it and take the right precautions (goggles, gloves, fume cupboard).

7C 6b IDENTIFYING HAZARDOUS ACIDS

In this experiment you will be comparing three different acids. You will find out which is the most hazardous.

Apparatus

- Test tube rack
- Test tubes
- Acids A, B and C
- 3 pieces of magnesium strip
- 3 marble chips
- Thermometer
- Eye protection



Eye protection should be worn.

Method

- 1 Fill a test tube about one third full of acid A, as shown in the diagram. Take the temperature of the acid and record it in the table.
- 2 Put **one** piece of magnesium into the tube.
- 3 Watch what happens. Write the results in the table.
- 4 When the fizzing stops, measure the temperature of the solution again.
- 5 Repeat steps **1-4** with acids B and C.
- 9 When the tests are finished, pour all the liquid away.
- 10 Refill the three tubes with fresh acid: A, B and C.
- 11 Do steps **1-4** again but using a marble chip instead of the magnesium.

Recording your results

Acid	Solid	Temperature at the beginning of the experiment (°C)	Observation	Temperature at the end of the experiment (°C)
A	magnesium		Fizzed quickly with lots of bubbles; the magnesium dissolved.	Rose (e.g. 20 °C to 45 °C)
B	magnesium		Fizzed slowly with a few bubbles.	Rose a little (e.g. 20 °C to 25 °C)
C	magnesium		No reaction - no bubbles.	Stayed the same (20 °C)
Acid	Solid	Temperature at the beginning (°C)	Observation	Temperature at the end of the experiment (°C)
A	marble chip		Fizzed steadily with bubbles; the marble chip slowly dissolved.	Rose (e.g. 20 °C to 35 °C)
B	marble chip		Slight fizzing and a few bubbles.	Rose slightly (e.g. 20 °C to 23 °C)
C	marble chip		No reaction - no bubbles.	Stayed the same (20 °C)

Considering your results

I think the most hazardous acid and the strongest acid was

Acid A - it reacted the fastest (most bubbles) and the temperature rose the most.

The least hazardous acid and the weakest acid was

Acid C - it reacted the slowest (hardly any bubbles) and the temperature barely changed.

I could tell this because The stronger the acid, the faster the reaction and the bigger the rise in temperature.

Suggest if the marble chips or the magnesium give us the most accurate way to tell which acid was the most hazardous or strongest acid? Explain why.

Marble chips - they react slowly and steadily so the time taken (or gas produced) can be measured accurately; magnesium reacts so quickly that comparing the acids fairly is difficult.

Extension

Was this a fair test? Explain.

No, it was not a fair test - the mass and size of the chips, the volume and concentration of the acid and the temperature must all be kept the same so that only the acid changes.

How could you make this practical fairer or more accurate?

Use the same mass and same sized chips (same surface area), the same volume and concentration of acid and the same temperature, and repeat each test to calculate an average.

What do we mean by 'strong' acid?

A strong acid is one that is completely ionised (dissociated) in water, releasing many hydrogen (H^+) ions, so it has a very low pH.

7C 7a Identifying Acids and Alkalis

Introduction: Litmus is a type of indicator. It goes red in acid, blue in alkali and stays purple in neutral substances.

Method:

1. Set up a dimple tray. 1. Set up a dimple tray on a clean, dry white tile.
2. Add a small amount of one of the substances into one of the dimples using a spatula or a plastic dropper pipette.
3. Add 1 drop of litmus indicator. Observe the colour and record this in your table.
4. Repeat for the other substances.

3. Add 1 drop of litmus indicator. Observe the colour and record it in the results table.
2. Add a small amount of one of the substances into one of the dimples.

Substance	Colour in Litmus	Acid, alkali or neutral?
Sulfuric acid	Red	Acid
Hydrochloric acid	Red	Acid
Vinegar	Red	Acid
Lemon juice	Red	Acid
Sodium hydroxide	Blue	Alkali
Soap	Blue	Alkali
Toothpaste	Blue	Alkali
Baking soda	Blue	Alkali
Salt	Purple	Neutral
Sugar	Purple	Neutral
Milk	Purple	Neutral

4. Repeat steps 2-3 for the other substances, using a clean stirring rod for each one.

Materials required

Sulfuric acid, toothpaste, vinegar, soap, salt, baking soda, sugar, lemon juice, sodium hydroxide, hydrochloric acid, milk, alcohol, dropping pipettes, dimple trays, litmus, phenolphthalein indicator.



Safety.

Wear goggles.
Treat all chemicals as corrosive and hazardous.
Wash your hands at the end of the practical.

Analysis/Conclusion :

- 1 Which of the chemicals are acids? Sulfuric acid, hydrochloric acid, vinegar and lemon juice.
- 2 Which of the chemicals are alkalis? Sodium hydroxide, soap, toothpaste and baking soda.
- 3 Which of the chemicals are neutral? Salt, sugar and milk.

Extension

Does litmus indicator tell us anything about how strong or weak our acids and alkalis are?

No - litmus only tells you whether a substance is an acid, an alkali or neutral; it does not show how strong or weak it is. Universal indicator (pH) is needed for that.

If you have time, try the same test with another indicator, such as phenolphthalein. What colour does this change in acid compared to alkali?

Phenolphthalein is colourless in acid (and in neutral) and pink in alkali.

7C 7b Acids Bingo - Review

Choose one of the words below to write into each box; make sure you put them in randomly.

Your teacher will read out a definition and you must cross out the matching word.

Try to be the first to get a) diagonal line, b) vertical line or c) horizontal line completed

Shout out 'bingo' when you do.

litmus	acid	sodium hydroxide	alkali
neutral	soap	vinegar	corrosive
hydrochloric acid	hazardous	irritant	litmus

These are the words you need to write into the boxes:

litmus

alkali

vinegar

hydrochloric acid

acid

neutral

corrosive

hazardous

sodium hydroxide

soap

litmus

irritant



MYP 1: Year 7 Science
Criterion D Assessment
Reflecting the Impacts of Science

UNIT:	Chemical Reactions
ASSESSMENT TITLE:	Acids and Alkalis in the Home
Statement of Inquiry:	Observing chemical reactions gives evidence to help understand patterns and relationships in the world around us.
Key Concept:	Relationships
Related Concepts:	Patterns and Evidence
ATL Focus:	Communication Skills
Key Scientific Vocabulary:	Hydrogen, Hydroxide, Corrosive, Irritant, Concentration, Neutralisation, pH Scale, Indicator, Dissociation, Salt, Solution, Solvent, Structure, Properties

GRASPS - ACIDS AND BASES IN THE HOME

There are many common acids and alkalis found in the home, examples include acids such as vinegar, fruit juice, lemonade, cola, ketchup, car batteries and alkalis such as soap, toothpaste, washing powder, oven cleaner and antacids.

How are we able to use the properties of specific acids and alkalis in our everyday lives? What are the implications in terms of safety compared to how useful they are?

Weigh the benefits of each chemical against its hazards: e.g. concentrated acid is very useful but corrosive, so it must be handled with care.

For this assignment, the setting is as follows...

Goal – your goal is to raise awareness of acids and bases in the home to make people realize that Chemistry is all around them.

Role – You work for the Royal Society of Chemistry, publicity division.

Audience – Youth and young adults, be sure to only use words that laymen (everyday people) would understand. Be sure to use correct scientific terminology when required.

Situation – You need to make a product that will encourage young people to take an interest in chemistry, while informing them of the science behind acids and bases and the prevalence of them in everyday lives.

Product – Make a leaflet or poster that will capture both minds and imaginations and encourage chemical thought! The product has to be informative, well researched and eye catching.

Illustrate and present your project using **media literacy (ICT)**.

Be sure to correctly use and reference a multitude of sources. Pictures also need to be referenced too. You will be assessed on using correct scientific terminology when required. Make sure you address your leaflet/poster to the target audience! Be as creative as you can but don't compromise on scientific

Other Information:

- Due Date – this will be set by your teacher and must be adhered to

Success Criteria for Assessment	For top marks, students must include: A leaflet/poster that has been created using ICT. At least 2 correctly referenced diagrams. At least 1 chemical equation showing how acids and bases interact with water. 2 properties each of an acid and an alkali and an explanation of how these can be identified. Specific examples (along the chemical formula) of 2 acids and 2 bases used in the home along with their purpose. A description of how the specific properties of both an acid and alkali help them with their function in the home. <i>“i.e. Sodium hydroxide is present in oven cleaner because...”</i> A summary of the health and safety implications and a discussion regarding the advantages and disadvantages of using these chemicals in the home. A clear and concise bibliography, with at least 5 different sources used.
---------------------------------	---

Level Descriptors and Task Specific Clarifications for Criterion D:

	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	No task specific clarifications addressed.
1-2	The student is able to, with limited success : i. state the ways in which science is used to address a specific problem or issue ii. state the implications of using science to solve a specific problem or issue, interacting with a factor iii. apply scientific language to communicate understanding iv. document sources	i. State where acids and bases are found the home. ii. State , briefly, the health precautions one should take when using acids and bases, with some inaccuracies. iii. Language used is casual or unclear throughout; points not made clearly. iv. limited sources used and cited.
3-4	The student is able to: i. Outline the ways in which science is used to address a specific problem or issue ii. Outline the implications of using science to solve a specific problem or issue, interacting with a factor iii. sometimes apply scientific language to communicate understanding iv. sometimes document sources correctly.	i. State , briefly, the basic properties of Acids and Bases. ii. State where acids and bases are found in the home and the safety precautions one should take when handling them. iii. Language used is sometimes clear; scientific terms sometimes applied. iv. Some sources used and cited.
5-6	The student is able to: i. Summarize the ways in which science is used to address a specific problem or issue ii. Describe the implications of using science to solve a specific problem or issue, interacting with a factor iii. usually apply scientific language to communicate understanding clearly and precisely document sources correctly.	i. Outline how the properties of acids and bases help them to be used for certain purposes around the home. ii. Outline the advantages and disadvantages of using acids and bases in the home and the precautions one should take when handling them. iii. Language used is usually clear; scientific terms generally used correctly. iv. Adequate sources used and cited; sources generally reliable.
7-8	The student is able to: i. Describe the ways in which science is applied and used to address a specific problem or issue ii. Describe and summarize the implications of using science and its application to solve a specific problem or issue, interacting with a factor iii. consistently apply scientific language to communicate understanding clearly and precisely iv. document sources completely	i. Describe how the properties of acids and bases help them to be used for certain purposes around the home, with specific examples of the chemical(s) used. ii. Describe and summarise the advantages and disadvantages of using acids and bases in the home; give reasoning to support the conclusion on whether acids/bases should be used. iii. Language used is usually clear; scientific terms generally used correctly.

iii. **Consistently apply** scientific language to communicate understanding **clearly and precisely**. Numerous reliable sources used and cited.

ii. State the implications of using science to solve a specific problem or issue.

iii. Apply scientific language to communicate understanding.

iv. Document sources (at a basic level, with support).

ii. Outline the implications of using science to solve a specific problem or issue.

iii. Sometimes apply scientific language to communicate understanding.

iv. Sometimes document sources correctly.

Student Task Planning Notes:

7C 10a Investigating Red Cabbage Indicator 1

Materials required

Red cabbage leaves, Pestle and mortar, Sand, Filter paper and funnels, Conical flasks & beakers
0.1M HCl, 0.1 M NaOH, Water bottles, Unknown substances A & B

These are the materials for the test: 0.1 M hydrochloric acid, 0.1 M sodium hydroxide, water bottles, and unknown substances A and B.

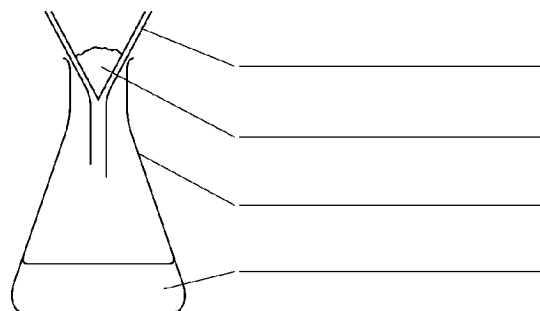
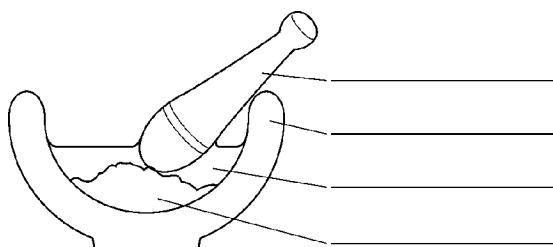


Safety. Wear goggles., Acids and alkalis are irritants/corrosive. Treat unknown substances as irritants.

Method/Diagram:

Your teacher will explain what to do:

To make sure you understand, before you start the experiment, label the following diagrams and fill in the missing words on the method below.



· Labelling the two diagrams: 1) pestle and mortar with sand + red cabbage leaves + water (grinding the cabbage); 2) filter funnel with filter paper in a conical flask, collecting the purple/dark red cabbage juice.

In this experiment the water is the solvent.

Some of the red colour dissolves in the water.

The rest of the cabbage leaves do not dissolve. They are insoluble.

The cabbage juice is separated from the leaves by filtering.

Dissolves,
filtering,
insoluble,
water

Results :

Name of Substance	Colour of Red Cabbage Juice
Acid (Hydrochloric)	Red (pink)
Neutral (Water)	Purple (violet)
Alkali (Sodium Hydroxide)	Green (yellow-green)
Unknown solution A	Red / pink - so unknown A is an acid
Unknown solution B	Green - so unknown B is an alkali

Analysis/Conclusion

The red cabbage juice changed red in acid, purple in neutral and green in alkali.

What type of substances were the unknown solutions A and B?

A = acid (indicator turns red) B = alkali (indicator turns green)

Describe in your own words what you think an indicator is:

An indicator is a dye/substance that changes colour to tell you whether a solution is acidic, alkaline or neutral.

Explain why you think scientists may find indicators useful:

Indicators are useful because they quickly and safely show whether a substance is an acid or an alkali (and roughly how strong it is), without having to taste or touch the chemical.

Evaluation

Why did you need to crush the red cabbage up before mixing it with water?

Crushing the cabbage breaks open the cells so the coloured dye inside dissolves in the water - it gives a bigger surface area and a much stronger colour.

How could you change your method to make your red cabbage indicator better?

Use less water (or more cabbage) so the colour is stronger, crush it for longer, leave it to soak, then filter carefully to remove all the bits of leaf.

Do you think red cabbage is a good indicator compared to litmus or others? Why?

Yes - it is cheap, safe and natural and gives different colours for acids, alkalis and neutral solutions; however litmus gives a clearer, standard colour change.

Extension

Different indicators change different colours. Find out the name of another indicator and what colour it is in acid, alkali and neutral conditions.

Phenolphthalein: colourless in acid, colourless in neutral, pink in alkali.
(Methyl orange is red in acid, orange in neutral and yellow in alkali.)

Find out how indicators work. What causes them to change colour?

Indicator dyes change their structure when the hydrogen ion (H^+) concentration changes, so they absorb and reflect different colours in acid and alkali.

7C 11a The pH Scale



Safety. Wear goggles.
Treat all chemicals as irritants.

Chemicals/Equipment

UI paper, glass rods, white tiles, several household substances .

Method:

Pick up a sample of one of the household substances. They may be in a beaker. Take this to your work bench. Write the name of the substance in the table.

- 2 Get some pieces of universal indicator (UI) paper and put these on a white tile.
- 3 Use a glass rod to get a drop of the household substance and put this onto the UI paper on the tile.
- 4 Write the colour into the table.
- 5 Look up the pH number using the colour chart. Write it into the table.
- 6 Do this again with another substance. Make sure you put the beakers of the substances back so that others can use them.

Results:

Name of substance	Colour of universal indicator	pH number	Type of substance
Hydrochloric acid	Red	1	Strong acid
Lemon juice	Red	2	Strong acid
Vinegar	Orange	3	Weak acid
Cola / fizzy drink	Orange	3	Weak acid
Milk	Yellow-green	6	Weak acid
Water	Green	7	Neutral
Baking soda solution	Blue-green	8	Weak alkali
Toothpaste	Blue	9	Weak alkali
Soap solution	Blue-purple	10	Weak alkali
Oven cleaner	Purple	13	Strong alkali

Considering your results/conclusion

Try to work out from the pH number if each of the substances you have tested is:

- a strong acid
- a weak acid
- neutral
- a weak alkali
- a strong alkali.

Write your answer in the column of the table marked 'Type of substance'.

Why do you think knowing the pH is useful?

Knowing the pH tells you how acidic or alkaline (and how strong) a substance is, so you know whether it is safe to use and how to treat it - e.g. in food, soil, medicines and cleaners.

Extension: Why can universal indicator change so many colours?

Can you find other examples of strong acids or alkalis that are commonly found in the home?
Predict their pH number.

Universal indicator is a mixture of several dyes, each one changing colour over a different pH range, so the mixture shows a different colour for every pH value.

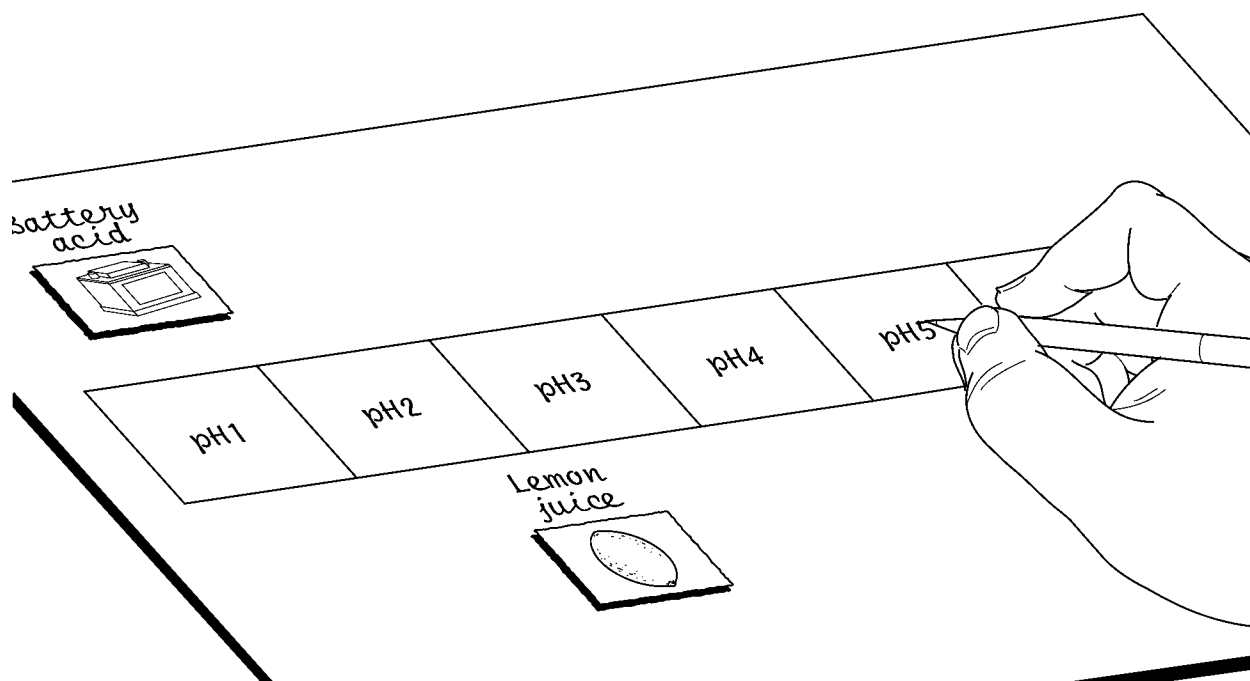
7C 11b Making a pH Scale

Use the blank sheet on the next page to draw the pH scale running from 1 to 14.

Add the colours to show how universal indicator would change colour.

Label the chart by drawing arrows from each substance to the correct pH number. You may be able to draw pictures or find photos that you could cut out and stick onto your chart.

Substance	pH	Substance	pH
battery acid	1.0	mineral water	8.0
bee sting	3.5	oven cleaner	13.0
blood	7.5	skin	5.5
distilled water	6.0	tap water	7.0
egg white	9.0	toothpaste	9.5
fizzy drinks	4.5	vinegar	3.0
indigestion powder	8.5	washing up liquid	5.0
kitchen surface cleaner	11.0	washing powder	10.5
lemon juice	2.5	washing soda	11.5
milk	6.5	wasp sting	10.0



Extension: Find out what 'pH' actually stands for: pH stands for 'potential of hydrogen' (power of hydrogen).

How is it calculated? It is calculated from the concentration of hydrogen ions in the solution: $\text{pH} = -\log_{10}[\text{H}^+]$.

What role do hydrogen ions play in pH?

pH scale chart: draw a scale from 1 to 14 with the universal indicator colours - 1-2 red, 3-4 orange, 5-6 yellow, 7 green, 8-9 blue, 10-11 dark blue, 12-14 purple - then draw arrows from each substance (battery acid 1.0, bee sting 3.5, blood 7.5, skin 5.5, distilled water 6.0, tap water 7.0, egg white 9.0, toothpaste 9.5, fizzy drinks 4.5, vinegar 3.0, indigestion powder 8.5, washing up liquid 5.0, kitchen surface cleaner 11.0, washing powder 10.5, lemon juice 2.5, washing soda 11.5, milk 6.5, wasp sting 10.0, mineral water 8.0, oven cleaner 13.0) to its pH number.

pH Scale

· Draw the pH scale 1 to 14 across the sheet with the universal indicator colours (red at 1 through orange, yellow, green at 7, blue, to purple at 14) and label arrows from each substance in the table on page 41 to its pH number.

7C 12a Changing Colours of Indicators 1

Aim: MAKING A RAINBOW BY NEUTRALISATION



Safety. Treat all chemicals as irritants at these concentrations, wear goggles and wash hands.

Chemicals/Equipment

Test tube rack containing six test tubes, 0.5M ethanoic acid, 0.5M NaOH, UI solution, measuring cylinders

Aim: In this activity you will also need to follow instructions carefully and also make precise observations. If you are careful, you should be able to produce a rainbow effect in the tubes.

Method:

- 1 Line the 6 test tubes up in the test tube rack.
- 2 Add about 5cm^3 of ethanoic acid to each using a measuring cylinder or syringe.
- 3 Carefully smell the contents of the test tubes by wafting them. Your teacher will explain how to do this safely.
- 4 Write your observations in the results section.
- 5 Add approx 3-4 drops of universal indicator to each of the test tubes. Carefully shake by tapping them gently with 2 fingers. Your teacher will show you how to mix chemicals safely.
- 6 What do they look like? What does this tell us? Write your observations in the results table.
- 7 Half fill a small beaker with the sodium hydroxide alkali solution (NaOH).
- 8 Leave the first test tube. Add nothing to this one.
- 9 Use a dropper pipette to add 5 drops of the NaOH alkali to the second test tube.
- 10 Gently tap the tube to shake it. Continue adding 5 drops of NaOH until the colour turns **orange**.
- 11 Write down the number of drops of NaOH added and the pH in the results table.
- 12 Do the same thing to test tube 3 until this one turns **yellow**.
- 13 In test tube 4 continuing adding 5 drops of NaOH until the test tube turns **green**. Feel the test tube.
- 14 Test tube 5 should turn **blue**.
- 15 Test tube 6 should turn **purple**.
- 16 Make sure you fill in your results and observations in the table

Results:

Boiling Tube	Colour in Universal	pH	Number of drops of NaOH (alkali)
1	Light Red	1 or 2	0
5 2		Orange	4
10 3		Yellow	6
15 4		Green	7
20 5		Blue	8 to 10
25 6	Purple	12 to 14	

Analysis/Conclusion:

- 1 What does the ethanoic acid smell like? vinegar (a sour, vinegary smell)
- 2 What colour does the ethanoic acid give when with universal indicator. red (light red)
3. What does this tell you about the pH? the pH is low - about pH 2 to 3 (acidic)
- 4 What type of a substance is ethanoic acid? an acid (a weak acid - it is only partly ionised)
- 5 Draw a diagram of your results (or take a photo and stick it in)
 - Diagram: six test tubes in a rack showing the rainbow - tube 1 light red, 2 orange, 3 yellow, 4 green, 5 blue, 6 purple - with the number of drops of NaOH added written underneath each tube.
7. NaOH is called sodium hydroxide. What type of substance is NaOH?
NaOH is an alkali - a soluble base (metal hydroxide) that produces hydroxide (OH-) ions in water.
8. What type of reaction is occurring? neutralisation
9. How many drops of NaOH are needed to make the acid neutral? about 15 drops (until the indicator turns green)
10. When you felt the test tube, did you feel any temperature change?
Yes - the test tube felt warmer. The reaction is exothermic: neutralisation releases heat energy to the surroundings.
11. What is also produced in most of these types of reactions
water (and a salt)

Extension: Find out other ways in which an acid can be neutralized.

An acid can also be neutralised by reacting it with a metal hydroxide, a metal oxide, a metal carbonate (e.g. baking soda) or a reactive metal.

What is a salt? Find the names of some common salts.

A salt is the compound formed when the hydrogen of an acid is replaced by a metal. Common salts: sodium chloride, potassium nitrate, copper sulfate, calcium chloride.

What is an alkali? Find the chemical names of a few.

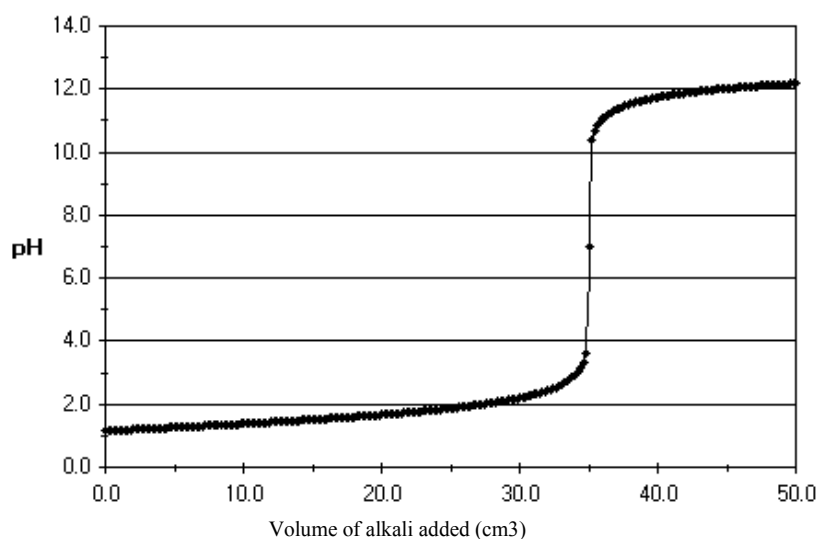
An alkali is a soluble base that produces hydroxide (OH-) ions in water. Examples: sodium hydroxide, potassium hydroxide, ammonia solution, calcium hydroxide.

What is an exothermic reaction?

An exothermic reaction is a reaction that releases energy (heat) to the surroundings, so the temperature of the mixture rises.

7C 12b pH Curves during Neutralisation

The graph below shows how pH changes as an alkali is slowly added to a beaker of acid.



Use the graph to answer the following questions;

- a) How can you tell from the graph that the beaker contained acid to begin with?

At 0 cm³ of alkali the pH is very low (about pH 1-2), which shows the beaker started with an acid.

- b) How much alkali had to be added to make the solution neutral? Explain how you made your decision.

About 30 cm³. I looked for where the curve rises steeply and passes through pH 7 - the neutral point - between pH 6 and pH 8.

- c) At the end of the experiment is the solution in the beaker acid, alkali or neutral? Explain how you decided.

Alkali. At the end the pH stays high (about pH 12-13) and does not drop back to 7, so there is excess alkali in the beaker.

- d) There was 20 cm³ of acid in the beaker at the beginning. Using the graph explain if the alkali added was stronger or weaker than the acid?

The alkali was weaker (more dilute): 20 cm³ of acid needed about 30 cm³ of alkali to reach pH 7, so more alkali than acid was needed.

- e) The beaker also contained universal indicator. Describe the colour you would observe at each of the volumes of alkali added:

0 cm³ = 0 cm³ = red (pH 1-2) 10 cm³ = 10 cm³ = orange (pH 3-4) 20 cm³ = 20 cm³ = yellow (pH 5-6)

30 cm³ = 30 cm³ = green (pH 7, neutral) 40 cm³ = 40 cm³ = blue (pH 9-10) 50 cm³ = 50 cm³ = purple (pH 12-13)

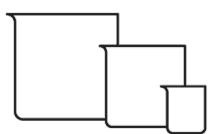
7C 13a INVESTIGATING ANTACIDS – Student Information Sheet

Which indigestion remedy works best?

Indigestion remedies react with acid in our stomachs and cancel it out (neutralize it). We can test different indigestion remedies by reacting them with some dilute acid and measuring the pH of the mixture.

Planning/designing your experiment

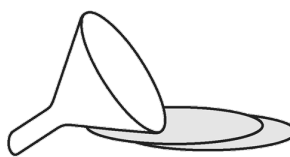
Equipment options



100cm³ beaker
250cm³ beaker
500cm³ beaker



Pestle and mortar



Funnel and filter paper



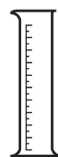
Hydrochloric acid
25cm³



Universal indicator
solution



Four types of antacid
tablets



Measuring cylinder



Other

VARIABLES

Control, independent

- | | |
|--|---|
| ☐ volume of acid | ☐ concentration of acid |
| ☐ type of antacid | ☐ amount of antacid powder added |
| ☐ amount of indicator | |

Dependent Dependent variable: the pH of the solution after 5 minutes reacting with the antacid (measured with universal indicator).

- ☐ pH of solution after 5 minutes reacting with antacid

SAFETY

Risks: hydrochloric acid is corrosive/irritating - wear goggles and wash off spills with water at once; do not eat the antacid tablets; glass beakers can break - handle carefully.

Risks

- | | |
|---|------------------------------------|
| ☐ hydrochloric acid | ☐ spillages |
| ☐ eating antacid tablets | ☐ burns |
| ☐ glass beakers | ☐ other? |

Test each remedy: the one that neutralises the most acid, or raises the pH closest to 7 the fastest, works best.

REPEATS

Will you repeat each measurement to make your data more reliable?

Yes - repeat every measurement at least three times and calculate an average, so the results are more reliable.

Recording your results

Design a results table to record your results.

· Results table columns: Brand of antacid | Volume of acid (cm³) | pH after 1, 2, 3, 4, 5 minutes | Average pH (with 3 trials for each brand).

You will need to draw a graph to present your results too.

Conclusion

What happened to the pH when the remedies reacted with the acid?

The pH rose (the acid was neutralised) - it moved from very low towards pH 7 because the antacid cancelled out the acid in the mixture.

Which remedy worked best? The remedy that made the pH rise fastest and closest to pH 7 (the one that neutralised the acid best).

How do you know? Give some examples from your graph.

My graph shows brand X rising steeply from pH 1 to pH 7 in about 2 minutes while brand Y flattened at only pH 4, so X worked best.

Try and explain what happened in terms of Science and neutralization.

Evaluation

Can you suggest any **errors** in your experiment?

If so, what **improvements** could you suggest to make the results more precise?

Improvements: repeat each test 3 times and average; use a pH meter for exact readings; use the same mass of fully crushed tablet; keep the acid volume, concentration and temperature the same.



MYP 1: Year 7 Science

Criterion B Assessment – Inquiring and Designing

Criterion C Assessment – Processing and Evaluating

UNIT:	Chemical Reactions	
ASSESSMENT TITLE:	Investigating Which Antacid is Most Effective at Neutralizing Stomach Acid	
Statement of Inquiry:	Observing chemical reactions gives evidence to help understand patterns and relationships in the world around us.	
Key Concept:	Relationships	
Related Concepts:	Patterns and Evidence	
ATL Focus:	Thinking Skills Social and Collaboration Skills	
Key Scientific Vocabulary:	Data	Measurement of a parameter that can be quantitative (volume, temperature, pH and so on) or qualitative (colour, shape, texture and so on).
	Dependent variable	The variable in which values are measured in the experiment.
	Prediction	Give an expected result of an upcoming action or event.
	Hypothesis	A tentative explanation for an observation or phenomenon that requires experimental confirmation; can take the form of a question or a statement.
	Independent variable	The variable that is selected and manipulated by the investigator in an experiment.
	Qualitative data	Refers to non-numerical data or information that is difficult to measure in a numerical way.
	Quantitative data	Refers to numerical measurements of the variables associated with the investigation.
	Validity of the method	Refers to whether the method allows for the collection of sufficient valid data to answer the question. This includes factors such as whether the measuring instrument measures what it is supposed to measure, the conditions of the experiment and the manipulation of variables (fair testing).
Command Terms:	Design	Produce a plan, simulation or model.
	Evaluate	Make an appraisal by weighing up the strengths and limitations.
	Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
	Analyze	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.

Assignment Instructions (GRASPS)

Goal	You are required to investigate either: 1. Which antacid brand is the most effective at combating indigestion problems. - OR 2. How antacids affect pH over time.
Role	You are a pharmacist either... 1. ...recommending a brand of antacid to your customer (if Goal 1 is selected) 2. ...explaining to your customer how long it will take for the antacid to work (if Goal 2 is selected).
Audience	A customer in your pharmacy.
Situation	You will work collaboratively in groups of twos or threes. You are required to design an organized practical investigation that will assist you in addressing the goal above. You may refer to the practical work that has been completed already in the Chemical Reactions unit. You must then perform the practical you have designed so that you can collect results that will help you determine the relationship the brand of antacid and its ability to neutralize an acid. You will be given the following chemicals and equipment to conduct your investigation: HCl, beakers, pestle and mortar, Funnel and filter paper, universal indicator, brands of antacid tablets, measuring cylinders.
Product	A scientific report should be completed to include a design, analysis and evaluation section. This report will be an electronic document that will be submitted via Google classroom for marking and will communicate the science of your investigation. You teacher will advise you of the due date for the report.
Success Criteria for Assessment	The final scientific report MUST include (see MYP Practical Report Checklist for Students in your booklet): - Title - Research Question - Background information - Hypothesis - Variables - Apparatus - Method - Safety considerations - Results table - Observations - Calculations - Graph - Conclusion - Strengths and Weaknesses - Improvements

2. How antacids affect pH over time - measure the pH at regular time intervals as the antacid reacts.

1. Which antacid brand is the most effective at neutralising stomach acid (raising the pH the fastest)?

1. Recommend a brand of antacid to your customer, using your results, if Goal 1 is met.

Level Descriptors and Task Specific Clarifications for Criterion B:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No research question stated. ii. No prediction stated. iii. Variables not identified. iv. Method and safety precautions not stated.
1-2	The student is able to: i. select a problem or question to be tested by a scientific investigation ii. select a testable prediction iii. state a variable iv. design a method with limited success .	i. A research question is stated but not relevant to the investigation. ii. A prediction with limited reference to acids and alkalis. iii. Independent, dependent and controlled variables identified but with inaccuracies. Method of data collection not indicated. iv. A method is presented but incomplete or not relevant to the investigation. Safety precautions not mentioned.
3-4	The student is able to: i. state a problem or question to be tested by a scientific investigation ii. outline a testable prediction iii. outline how to manipulate the variables, and state how data will be collected iv. design a safe method in which he or she selects materials and equipment .	i. A relevant research question is stated, but is not too clear or focused. ii. A prediction with some reference to the variables is made. iii. Independent, dependent and controlled variables identified but incomplete. Basic method of data collection is stated. iv. Method is presented in steps but does not clearly show how the experiment is conducted and/or how data is collected. Safety precautions stated but are not specific to the experiment.
5-6	The student is able to: i. outline a problem or question to be tested by a scientific investigation ii. outline and explain a testable hypothesis using scientific reasoning . iii. outline how to manipulate the variables, and outline how sufficient relevant data will be collected iv. design a complete and safe method in which he or she selects appropriate materials and equipment .	i. A research question is stated with reference to some correct variables. ii. A prediction with reference to the relationship between the independent and dependent variables is outlined. iii. Independent, dependent and controlled variables correctly identified, with an outline of how relevant data is collected. iv. Method is presented in clear steps with relevant details, with suitable materials and apparatus chosen. A labelled diagram to show how the apparatus is set up. Relevant safety precautions

i. No research question stated - lowest band, no marks for the research question.

ii. No prediction stated - lowest band, no marks for the prediction.

7-8	The student is able to: i. describe a problem or question to be tested by a scientific investigation ii. outline and explain a testable hypothesis using correct scientific reasoning iii. describe how to manipulate the variables, and describe how sufficient, relevant data will be collected iv. Design a logical, complete and safe method in which he or she selects appropriate materials and equipment.	i. A research question clearly and explicitly identifies the independent and dependent variables with reference ii. A prediction with reference to the relationship between the independent and dependent variables is outlined and explained using relevant scientific reasoning. The explanation should mention neutralisation. iii. Independent, dependent and controlled variables correctly identified, outlining how they are manipulated, measured or controlled. Values of variables stated where appropriate. Outline how sufficient, relevant data is collected. iv. A complete method is presented in clear steps with all relevant details, stating values and measuring equipment used. A fully labelled diagram to show how the apparatus is set up. v. Safety precautions discussed and are relevant to this investigation.
-----	--	--

Level Descriptors and Task Specific Clarifications for Criterion C:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No data presented. No graph plotted. ii. Trend not stated. iii. No conclusion on whether the prediction is supported by data. iv. No strengths or limitations relevant to the investigation stated.
1-2	The student is able to: i. collect and present data in numerical and/or visual forms ii. Accurately interpret data iii. state the validity of a hypothesis with limited reference to a scientific investigation. iv. state the validity of the method with limited reference to a scientific investigation v. state limited improvements or extensions to the method	i. Raw, quantitative data is presented in a table. A graph is attempted. ii. A trend is stated but is not supported by data. No scientific explanation given to explain the trend. iii. A conclusion is attempted, but with inaccuracies. iv. At least one limitation relating to the investigation is stated, but not relevant to the investigation. v. At least one improvement or extension stated but is not relevant to the limitation identified.
3-4	The student is able to: i. correctly collect and present data in numerical and/or visual forms ii. accurately interpret data and describe results	i. Raw, quantitative data is neatly presented in a table with headings. A graph is plotted with labelled axes. ii. Trend shown in the graph is outlined by stating a basic relationship between variables.

	iii. state the validity of a hypothesis based on the outcome of a scientific investigation iv. state the validity of the method based on the outcome of a scientific investigation v. state improvements or extensions to the method that would benefit the scientific investigation.	iii. A correct conclusion is drawn with reference to the prediction. iv. At least one limitation relating to the validity of the investigation is stated. v. At least one improvement relating to each limitation raised is stated.
5-6	The student is able to: i. correctly collect, organize and present data in numerical and/or visual forms ii. accurately interpret data and describe results using scientific reasoning iii. outline the validity of a hypothesis based on the outcome of a scientific investigation iv. outline the validity of the method based on the outcome of a scientific investigation v. outline improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data from multiple trials are neatly presented in a table with headings and relevant units. An attempt has been made to calculate averages. A correct type of graph is plotted. The graph has correctly labelled axes including units and suitable scales. Some observations are recorded. ii. Trend shown in the graph is correctly outlined and explained using some correct scientific knowledge relating to neutralisation. Anomalies/outliers are identified. iii. A conclusion is drawn. Student states if the data supports the prediction. iv. At least two limitations relating to the validity of the investigation are outlined. v. At least one improvement relating to each limitation is outlined. Extensions of investigation outlined. iii. A conclusion is drawn. Student states if the hypothesis is supported by the data.
7-8	The student is able to: i. correctly collect, organize, transform and present data in numerical and/ or visual forms ii. accurately interpret data and describe results using correct scientific reasoning iii. discuss the validity of a hypothesis based on the outcome of a scientific investigation iv. discuss the validity of the method based on the outcome of a scientific investigation v. describe improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data from multiple trials are neatly presented in a table with informative headings and relevant units. Averages are correctly calculated and presented in the table. A correct type of graph is accurately plotted. The graph has a title, correctly labelled axes including units and suitable scales. A line of best fit is included. Meaningful observations are recorded. ii. Trend shown in the graph relating to the factor investigated is correctly outlined with reference to specific data points. Trend is explained using correct scientific knowledge relating to neutralisation. Anomalies/outliers are correctly identified. iii. Correct conclusion is drawn as to how well the data supports the prediction, with reliability of the data discussed. iv. One strength and three limitations relating to the validity of the investigation are discussed. Consider how the limitations affected the data and how well the research question could be answered. v. At least one improvement relating to each limitation raised is described. Relevant extensions of investigation described. iii. A conclusion is drawn. Student states if the hypothesis is supported by the data.

Student Task Planning Notes:

Planning notes should cover: research question, hypothesis, independent/dependent/controlled variables, apparatus list, step-by-step method, safety precautions, a results table with headings and units, and how the graph will be plotted (pH against time/brand).



MYP Science Practical Report Template

Name:

Form:

UNIT: Chemical Reactions**Statement of Inquiry:****Key Concept:** Relationships**Related Concepts:** Patterns and Evidence

Investigation Title: Investigating Which Antacid is Most Effective at Neutralising Stomach Acid
 Observing chemical reactions gives evidence to help understand patterns and relationships in the world around us.

Criterion B - Inquiring and Designing

Research Question

(What are you trying to find out?)

Research question: Which brand of antacid tablet is the most effective at neutralising stomach acid (hydrochloric acid)?

Background Scientific Information and Hypothesis

(What do you think will happen and why?)

Hypothesis: the brand containing the most antacid/carbonate will neutralise the acid fastest and raise the pH closest to 7, because more alkali means more acid is neutralised.

Variables

Independent <i>(What will change and how will you change it?)</i>	Independent variable - the brand/type of antacid tablet used. I will change the brand (four different brands) keeping everything else the same.
Dependent <i>(What will you measure to gather result?)</i>	Dependent variable - the pH of the mixture, measured with universal indicator solution (or UI paper/pH meter) at the same times after the tablet is added.
Control <i>(What variables will you keep the same to make it a fair test? How will you keep the variables constant?)</i>	Keep the same: volume (25 cm ³) and concentration of hydrochloric acid, mass of antacid, amount of indicator, temperature and the time of each reading (5 minutes).

Apparatus Set-up

(Draw a labelled diagram of the assembled apparatus that you will use. Write the name of any equipment not shown in the diagram)

· Diagram: labelled 100 cm³ beaker of hydrochloric acid with a few drops of universal indicator on a white tile, plus the pestle and mortar, filter funnel and filter paper, and a measuring cylinder of acid.

Method

(Write a step by step account of how you conduct your experiment – the method must be DETAILED)

- 1.
 - 2.
 - 3.
 - 4.
 - 5.
1. Measure 25 cm³ of hydrochloric acid with a measuring cylinder and pour it into a 100 cm³ beaker.
 2. Add 5 drops of universal indicator to the acid and record the starting colour and pH.
 3. Crush one antacid tablet with the pestle and mortar and add the powder to the acid; start the stopwatch.
 4. Record the pH (indicator colour) every minute for 5 minutes, using the same colour chart each time.
 5. Repeat steps 1-4 for each of the other three brands, doing each brand three times, then calculate an average.

Safety

(How will you make sure your method is safe?)

Wear goggles and handle the acid (corrosive/irritant) carefully; wash off any spills with plenty of water; never eat the antacid tablets; carry glassware carefully and report breakages.

Reliability

(What will you do to ensure your results are reliable?)

Repeat each brand at least three times and average the results, control all other variables, take readings at exactly the same times, and use a pH meter or the same colour chart.

Criterion C - Processing and Evaluating Results

Raw Data

(A neat, organized table of results should be drawn, with headings, units, repeats and averages include)

Ruled results table: headings with units, e.g. 'Time after mixing (s)' 0/30/60/90/120 and 'Temperature of mixture (degrees C)' for repeat 1, repeat 2, repeat 3 plus a mean (average) column, values to the same precision.

Observations

(What changes did you notice as the experiment was conducted, e.g. colour change, bubbles, heat change?)

Start: colourless solution, no gas, no temperature change. During: bubbles of gas given off steadily, mixture became warm (temperature rose), solid shrank. End: solid had dissolved, bubbling stopped, no further change.

Graphing Results

You should draw a graph of your results and attach it to this report.

Analysis of Graph

(What did your results show? Are there any trends or patterns in your results? Can you give an example of a pattern or trend from your results?)

· Graph must be attached and hand-drawn: axes labelled with quantity and unit (e.g. time / s, temperature / degrees C), suitable scale (at least 3/4 of each axis), all points plotted, line or curve of best fit.

Positive trend: as the independent variable increases the dependent variable rises - steepest at the start, then flattening off (e.g. 20 degrees C at 0 s, 30 degrees C at 60 s, 33 degrees C at 120 s).

Conclusion and Evaluation

Scientific Conclusion and Explanation of Results

(Summarize what your results show. Does this support your prediction? Can you explain your results using scientific knowledge?)

My results show that as time increased the temperature also increased, which supports my prediction. This is because the reaction is exothermic - the reactants release energy to the surroundings as heat.

Evaluation of Scientific Method and Results

How reliable were your results? How well did your investigation answer the research question? What went well with your method?)

Results were reliable because repeats agreed closely, and the method answered the research question because it measured the correct dependent variable. What went well: the method was followed carefully and safely.

(Were there any problems or parts of your investigation that were not very accurate or could have been done better?)

Weakness: readings were taken by eye and the container was

Issue with justification	Suggested SCIENTIFIC Improvement
Heat was lost to the surroundings because the container was not insulated, so the measured temperature change was smaller than the true value.	Improvements: lag/insulate the container to reduce heat loss; take 3 repeats and calculate a mean; use a digital thermometer or data logger for accuracy; make sure every trial starts at the same temperature.
Only one set of readings was taken, so random errors could not be spotted and the reliability of the result is low.	
The thermometer was read by eye and had 1 degree C graduations, so each reading could be inaccurate by about +/- 1 degree C.	
The starting temperature of each trial was not exactly the same, so the temperature change of different trials could not be compared fairly.	

MYP Practical Report Checklist - Student

Before you submit your practical report to your teacher you must use the checklist below to ensure all parts of your report have been completed.

	Section	What you need to do	Completed Y/N
DESIGN (MYP Criterion B: Inquiring and Designing)	Title	Give a name to the experiment ✓ you are conducting	
	Research Question	The research question should be clear, concise , and include the independent variable and the dependent variable.	
	Background Information ☒	You should give the science theory behind your investigation, and it should include scientific terminology. This should be presented as a short scientific paragraph .	
	☒ Hypothesis	What do you predict will happen in your experiment and why ?	
	Variables	Independent Variable - what will you change and how will you change it? Dependent Variable - what will you measure and how will you measure it? Control Variables - what are they? How and why are they controlled? how will you measure it? State the instrument and units, e.g. pH meter for pH or stopwatch in seconds for time.	
	Apparatus	Have ✓ you completed a diagram of assembled equipment that shows how to successfully collect data? Have you used a ruler and pencil to do this? Have you labeled all the chemicals and equipment in your diagram? Have you used a ruler and pencil to do this? Yes - draw tables, axes and lines in pencil using a ruler.	
	Method	Have ✓ you written your method in the following format: - Stepwise - Logical order of steps in investigation - Used correct names for chemical and equipment - Provided detailed account of the steps in the method - Describe how you will change the independent variable - Describe how control variables are monitored	
	✓ Safety	Discuss any safety issues with chemicals or equipment being used, and describe what precautions should be taken to ensure you conduct the investigation safely.	

RESULTS and CONCLUSION (MYP Criterion C: Processing and Evaluating)	Results Table	A table of results must include: - Headings - Units - Repeats - Averages	
	Observations	Record detailed observations: - At start of experiment - During experiment - At end of experiment Observations can include colour change, bubbles, heat etc.	
	Calculations	Calculate averages of repeats. Show formula for calculation Show working out for calculations Give final answer to calculation with correct units .	
	Graph	Draw a graph to include the following; - Heading, labels with units on each axis - Suitable scales that occupy minimum $\frac{3}{4}$ of each axis - Plot all points accurately - Draw a curve or straight line of best fit that ignores points that do not fit the pattern. - Discuss trends or patterns observed in graph.	
	Conclusion	Summarize what your results show. Does this support your prediction/hypothesis ? Explain your results using scientific knowledge ?	
	Evaluation	Discuss the reliability of your results . Discuss the validity of the method you used. Explain your results using scientific knowledge, e.g. acid + alkali -> salt + water (neutralisation) explains the pH change.	
	Improvements	Suggest improvements to the scientific method and describe how they can reduce or eliminate errors	

Does this support your prediction/hypothesis? State yes/no and quote evidence from the results to justify it.

7C 15a Summary Sheets Chemical Reactions

Chemical reactions

In a **chemical reaction** a new substance is always formed. Most chemical changes are not easily reversed; they are **irreversible**. The chemicals that you start with are called the **reactants**. The chemicals at the end are called the products.

In a **physical change** no new substance is formed. Melting and evaporation are examples of physical changes. Physical changes are usually reversible.

You can tell that a reaction has occurred if there is a **colour change** or when a **gas** is given off. Most chemical reactions also involve an **energy change**. This is usually in the form of heat, but can also involve light being given off (for example, when something burns).

Reactions of acids

Some metals react with acids, and hydrogen gas is produced.

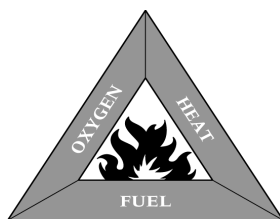
When acids react with chemicals called carbonates, carbon dioxide gas is given off. Carbonates are found in rocks such as limestone or marble, and in some cooking ingredients and indigestion tablets.

You can test the gas made in a reaction to find out what it is:

- **Hydrogen** burns with a squeaky pop if a lighted splint is held near the test tube.
- **Carbon dioxide** will put out a lighted splint, and it makes limewater turn milky.

Burning

A fire needs three things to keep burning: fuel, oxygen and heat. We show these three things on the **Fire Triangle**.



If any one of these three things runs out, the fire will go out.

When a metal burns, the metal combines with oxygen from the air to form a chemical called an **oxide**. We can show this using a **word equation**.

Combustion

Fossil fuels contain a lot of carbon and hydrogen. When they burn they use up oxygen from the air and produce water and carbon dioxide. We can show the reaction using a word equation. Energy is in brackets in this equation because it is not a chemical substance.

fuel + oxygen → carbon dioxide + water

Indicators are coloured dyes which often come from plants such as red cabbage and beetroot.

Acids make indicators change colour.

Litmus is an indicator which turns red in acids. Common acids include vinegar and lemon juice. Fizzy drinks, pickles and spicy sauces also contain acids. Stronger acids such as sulphuric and nitric acids can be more dangerous. Often they are **corrosive**.

Alkalis have a different effect on indicators to acids. Litmus turns blue in alkalis. Alkalis can also be corrosive. Weak alkalis include soap and toothpaste.

Bottles in the laboratory and tankers carrying chemicals on the road all have to carry hazard warning labels to show when there is a chemical hazard. Some of the common warning signs are:



toxic (poisonous)



harmful (irritant)



corrosive



flammable

The strengths of acids and alkalis can be measured on the **pH scale**, which runs from 1 to 14. pH numbers **1 to 6** are acids, **7** is neutral, and **8 to 14** are alkalis. You can find out the pH number using a **universal indicator**, or by using a pH meter.

strong acid			weak acid			neutral	weak alkali			strong alkali			
1	2	3	4	5	6	7	8	9	10	11	12	13	14

stomach
acid

vinegar

skin

pure
water

indigestion
powders

washing
powder

oven
cleaner

lemon
juice

fizzy
drinks

milk

blood

toothpaste

Alkalis can react with acids, making a **neutral** chemical.

Neutralization reactions can be important:

- in gardening and agriculture, to make sure the soil is the correct pH
- when dealing with insect stings and bites
- to control indigestion caused by excess acid in the stomach
- to keep foods such as jam at the correct pH.

7C 15b Quick Quiz (Chemical Reactions)

Circle the correct letter for each question.

7C 1

- 1 Two liquids were mixed together in a test tube. Which one of these would **not** show that a chemical reaction had taken place?
- A The tube got warm.
 - B Bubbles of gas were given off.
 - C A brightly coloured solid formed in the tube.
 - D One of the liquids floated on top of the other.
- 2 Which of these changes is a chemical reaction?
- A melting some margarine in a pan
 - B boiling a kettle of water
 - C frying an egg
 - D cooling a drink by putting an ice cube in it
- 3 The substances that you get at the end of a chemical reaction are called:
- A reactants.
 - B chemicals.
 - C products
 - D gases
- 4 A chemical reaction:
- A always produces a gas.
 - B is always reversible.
 - C makes new substances.
 - D does not make new substances.

7C 2

- 1 Which kind of chemical will react with an acid to form hydrogen?
- A metal
 - B water
 - C carbon
 - D petrol
- 2 To test for hydrogen you must:
- A Put the gas over a Bunsen Burner.
 - B Mix the gas with limewater.
 - C Put a lit splint into the gas.
 - D Smell the gas.
- 3 Hydrogen gas:
- A burns with a squeaky pop.
 - B makes a glowing splint relight and burn brightly.
 - C puts a glowing splint out.
 - D turns universal indicator solution red.
- reaction had taken place? Evidence includes fizzing/gas given off, colour change, temperature change or a new solid forming.

7C 3

- 1 Which two chemicals would you mix together to make carbon dioxide?
- A an acid and a metal
 - B an acid and a carbonate
 - C an alkali and a metal
 - D an alkali and a carbonate
- 2 Carbon dioxide is used:
- A to make the bubbles in fizzy drinks.
 - B to fill party balloons.
 - C to help people breathe more easily.
 - D in gas cookers, to provide heat.
- 3 Which of these will show that a gas is carbon dioxide?
- A milk turns to water
 - B rosewater turns pink
 - C limewater turns milky
 - D burns with a squeaky pop
- 4 Which substance does not contain carbonates?
- A indigestion tablets
 - B limestone
 - C baking powder
 - D sugar

7C 4

- 1 When a metal burns it:
- A combines with oxygen from the air.
 - B combines with nitrogen from the air.
 - C gives off carbon dioxide.
 - D gives off oxygen.
- 2 Which is the correct equation to show what happens when magnesium burns?
- A magnesium + nitrogen → magnesium nitride
 - B magnesium + oxygen → magnesium oxide
 - C magnesium + oxygen → carbon dioxide + water
 - D magnesium oxide → magnesium + oxygen
- 3 A fire needs three things to keep burning. They are:
- A coal, air, smoke
 - B natural gas, wood, heat
 - C heat, fuel, nitrogen
 - D heat, fuel, oxygen
- 4 A fire can be put out by:
- A removing its fuel.
 - B removing its oxygen.
 - C cooling it down.
 - D any of the above.

An acid and a carbonate, e.g. hydrochloric acid + calcium carbonate (marble chips).

Bubble the gas through limewater - it turns milky/cloudy, showing the gas is carbon dioxide.

The substance that does not fizz with dilute acid, e.g. water/sugar, because it does not contain carbonate.

7C 5

1 Which list contains only fuels? The list in which every substance burns to release energy, e.g. coal, oil and natural gas (hydrocarbons).

- A wood, iron, plastic
- B wood, petrol, oxygen
- C petrol, oil, natural gas
- D air, wood, oil

2 When a fuel burns there is:

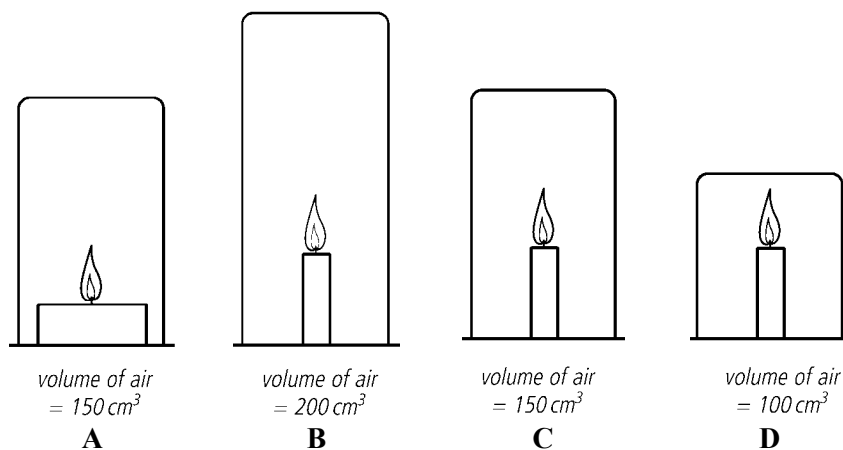
- A a reversible physical change.
- B a reversible chemical change.
- C an irreversible physical change.
- D an irreversible chemical change.

3 Which statement finishes off this word equation for burning?

fuel + oxygen → carbon dioxide + water (+ energy)

- A carbon dioxide + water (+ energy)
- B carbon dioxide + carbon monoxide
- C fuel oxide + oxygen (+ energy)
- D methane + carbon dioxide (+ energy)

4 Which of these candles will burn the longest?

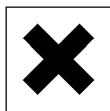


· Picture question: the candle that burns longest is the one with the largest supply of oxygen (the candle left open / not sealed under a container) - ring that picture.

7C 6

1 What does this sign mean?

- A flammable
- B harmful
- C poisonous
- D explosive



2 Acids:

- A are always dangerous.
- B are more dangerous when strong than when weak.
- C are more dangerous when weak than when strong.
- D can be made safe by heating them.

3 The hazard warning signs on chemical tankers:

- A tell the driver of the lorry what the maximum safe driving speed is.
- B tell other drivers the distance to stay back from the lorry.
- C allow the police to check the speed of the tanker.
- D help the emergency services deal with a spillage if there is an accident.

4 Alkalis:

- A are much safer than acids.
- B can be more dangerous than acids.
- C are always more dangerous than acids.
- D are not allowed to be transported by road.

7C 7

1 Acids taste:

- A sweet.
- B bitter.
- C sharp.
- D salty.

2 The name of an acid in lemon juice is:

- A citric acid.
- B hydrochloric acid.
- C sulphuric acid.
- D acetic acid.

3 Which of these foods will contain most acid?

- A frozen peas
- B rice pudding
- C pickled onions
- D chips

3. The food with the lowest pH (most acidic), e.g. lemon juice/citrus fruit or vinegar.

4 Which list has three foods which all contain vitamin C (ascorbic acid)?

- A peas, oranges, cabbage
- B bread, peas, baked beans
- C oranges, bread, baked beans
- D oranges, bread, cabbage

· Picture question (hazard sign printed in the figure): ring the option that matches the sign shown - the harmful/irritant (X) symbol, i.e. 'harmful'.

7C 8

1 Litmus can be used as an indicator because it:

- A is an acid.
- B is an alkali.
- C changes colour
- D comes from a plant.

2 Which one of these is an acid?

The one with a pH below 7 that turns blue litmus red, e.g. lemon juice or dilute hydrochloric acid.

- A tap water
- B salt water
- C vinegar
- D soap

3 Which one of these is an alkali?

The one with a pH above 7 that turns red litmus blue, e.g. sodium hydroxide or soap solution.

- A lemon juice
- B toothpaste
- C vinegar
- D tap water

4 Which of these is not neutral?

The substance with a pH that is not 7, i.e. an acid or an alkali, because neutral substances have pH 7.

- A salty water
- B sugary water
- C tap water
- D soapy water

7C 9

1 The pH scale indicates

- A how hot something is.
- B how acid or alkaline something is.
- C how long something is.
- D how much substance is dissolved in a solution.

2 The pH number of a strong acid is:

- A 1
- B 5.5
- C 7
- D 9

3 The pH number of a weak alkali is:

- A 1
- B 5.5
- C 7
- D 9

4 What colour will universal indicator turn when mixed with a strong alkali?

- A red
- B orange
- C green
- D purple

7C 10

- 1 A wasp sting has a pH of about 10. You could treat it best using:
- A a strong acid.
 - B a weak acid.
 - C a weak alkali.
 - D a strong alkali.
- 2 The word which best describes what happens when an acid reacts with an alkali is:
- A acidification.
 - B evaporation.
 - C neutralisation.
 - D alkification.
- 3 If you took a solution of pH 5 and diluted it with water, the pH number would:
- A go down by one or two.
 - B stay the same.
 - C go up by one or two.
 - D go up by three or four.
- 4 If you mixed a solution of pH 4 with one of pH 8, the final pH could be:
- A 2
 - B 6
 - C 10
 - D 12

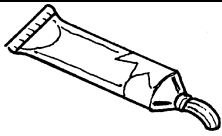
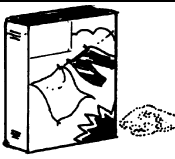
7C 16a Additional Practice Questions

CHEMICALS IN THE HOME

The chart shows the names of some substances you find in the home.

Five of them are acids, five are alkali. Use the internet to find out which are which. Write 'acid' or 'alkali' in each blank box. One has been done for you.

Acids (5): vinegar, picked onions, orange juice, lemonade, hair perming lotion. Alkalis (5): toothpaste, washing powder, soap, bleach, milk of magnesia.

				
vinegar	toothpaste	picked onions	orange juice	washing powder
acid	alkali	acid	acid	alkali

				
soap	lemonade	bleach	milk of magnesia	hair perming lotion
alkali	acid	alkali acid	alkali	acid

Extension: Find out what the word 'Alkali' means.

An alkali is a base (such as a metal hydroxide) that dissolves in water. Alkalis neutralise acids and have a pH between 8 and 14.

What can alkalis be used for?

Alkalis are used in toothpaste, antacid indigestion medicines, soap, bleach and washing powders, and to neutralise acids - e.g. acidic soil in farming or acid spills.

7C 16b CRACKING THE HAZCHEM CODE (Extension)

The number code:			
1	Jets	3	Foam
2	Fog	4	Dry agent
P	V	FULL	DILUTE
R			
S	V	BA	CONTAIN
T			
W	V	FULL	
X			
Y	V	BA	
Z			
E consider evacuation			

What the numbers mean:

- 1 Jets** means that the fire can be put out using jets of water from a hose.
- 2 Fog** means that the fire brigade can still use water. It has to be a fine mist or spray rather than full power jets.
- 3 Foam** means that the tanker needs to be sprayed with a layer of thick foam to smother the flames.
- 4 Dry agent** means that water must NOT be used on this fire. It has to be kept dry. Examples of dry agents are sand, powders, or carbon dioxide gas.

Using the information above attempt to answer the following questions



A



B



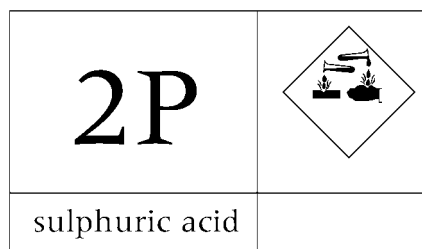
C



D

- Which of these hazard warnings means that a chemical is:
 - flammable D (flammable)
 - poisonous A (poisonous/toxic)
 - corrosive C (corrosive)
 - an irritant? B (irritant/harmful)

2 The Hazchem Code tells the police and fire services about the hazards of a chemical in a tanker. Look at the Hazchem symbol.



Hazchem code on a tanker.

- a What chemical is being carried in the tanker? **sulphuric acid**
- b What is the **number** on the Hazchem warning sign? **2**
- c What word matches this number on the code chart? **Fog**
- d How should the fire brigade put out a fire on this tanker? Choose from:
- A use jets of water
 - B use a fine spray mist of water
 - C use a dry powder.
- e What is the **letter** on the Hazchem warning sign? **R**
- f The 'V' means that there could be a violent reaction with the chemical. Is this a problem with the sulphuric acid?
No - the sign does not include a V, so a violent reaction is not listed as a hazard for this acid and a fine water spray can still be used.
- g The fire brigade will either have to wear breathing apparatus and gloves (BA) or a FULL protective suit. Which should they wear in this case?
BA - breathing apparatus and gloves (the code calls for BA rather than a FULL protective suit).
- h A spillage of the chemical may be able to be washed away (DILUTE). Some chemicals need to be collected up (CONTAIN). Which should the fire brigade do in this case?
DILUTE - the spillage can be washed away with a fine water spray (letter R = DILUTE).

- i Does the warning sign have an 'E' in the code? No
- j Should the police clear the area, or is it safe for people to stay where they are?
It is safe for people to stay where they are - no evacuation is needed.
- k How do you know?
Because there is no letter E in the code; E would mean 'consider evacuation', so without it the area does not need to be cleared.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p15: Add dilute acid to each material: carbonates fizz and give off carbon dioxide, which turns limewater milky/cloudy.

p18: Advantages: hydrogen burns to make only water, so it is a clean fuel. Disadvantages: it is explosive and hard to store. CO₂ extinguishers keep oxygen away from the fire.

p49: The equipment marked '?' is the measuring cylinder needed to measure out 25 cm³ of hydrochloric acid.

p50: The antacid is a base/carbonate, so it neutralised the hydrochloric acid, forming a salt and water; as the acid was used up the pH rose towards 7.

p50: Errors: only one trial was done; the pH was judged by eye from a colour chart; the tablets may not have dissolved fully; the tablet masses differed; temperature changed.

p70: 2. magnesium + oxygen -> magnesium oxide (word equation); $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$.